

BrickAtlas: Measuring Visual, Structural, and Executable Reasoning on Shared Brick Assemblies

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Abstract

Recognizing a brick assembly, reconstructing its hidden parts, modifying it, and specifying an executable build order are different problems. Their scores become misleading when input privileges, acceptable alternative structures, or evaluator interventions are left implicit. We present BrickAtlas, a controlled benchmark resource that connects eight task families through 5,120 procedural source assemblies and 117,910 task-condition cases. Explicit contracts separate ordinary RGB, layer disclosure, and privileged symbolic references, and distinguish visual compatibility from full structural recovery and plan execution. We introduce a source-level coverage audit and evaluate 35,840 copy-current controls: doing nothing obtains 87.1% part F1 on completion and 97.4% on removal, while solving neither task. A height-sorted planner solves all 10,240 planning cases, delimiting this domain as constraint following rather than difficult search. An exploratory two-model study on 24 assemblies exposes reconstruction floors and editing ceilings. Four-model candidate diagnostics on 12 additional development sources show more pose inconsistency under re-ordering than identical repetition. We release reproducible contracts, case-level outputs, figures, and audits, and explicitly separate these findings from the still-unmeasured independent confirmation, external-domain, and human-calibration experiments.

1. Introduction

What does it mean for a vision–language model to understand an assembly? A correct part list does not locate the parts. A matching exterior need not determine the interior. A high similarity score after an edit may simply preserve what was already correct. Even an accurate final configuration does not certify that a submitted construction sequence can execute. These distinctions matter when visual reasoning systems are evaluated as if recognition, structured prediction, and action planning were interchangeable.

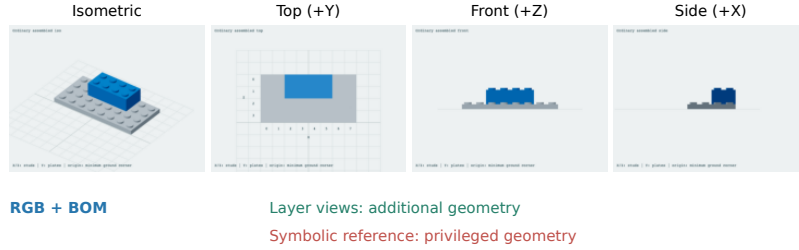
Brick assemblies make this problem concrete. They have discrete identities, poses, contacts, and operations, enabling explicit evaluation. At the same time, occlusion and repeated interchangeable components make one reference program an incomplete definition of correctness. A benchmark must specify both *what information was available* and *which answers that information justifies*. Figure 1 illustrates our evaluation contract.

Existing work provides strong generation and assembly capabilities: BrickGPT produces stable, buildable structures from text [6]; BrickNet learns graph-backed programs from human-designed LDraw assemblies [4]; MEP-Net parses visual manuals into component poses [10]; and Break and Make studies interactive structural understanding [9]. BrickAtlas does not replace their richer geometry, learned methods, or interactive settings. Its focus is a *measurement problem*: which conclusions survive when observations, answer equivalences, and execution are accounted for explicitly on shared objects?

We organize the study around three questions. **RQ1:** When does a high global structure score fail to indicate that the requested operation was completed? **RQ2:** Does visual compatibility determine downstream structural relations? **RQ3:** Are candidate-selection outcomes stable to a change in presentation, beyond instability under an identical repeated request? Each has an explicit negative outcome: similarity may agree with task success, admissible layouts may agree on a query, or permutation may be no less stable than repetition. No preferred model ranking is assumed.

Our contributions are threefold. First, we provide an eight-family casebank with source-group splits, typed answers, information conditions, and distinct visual, structural, and executable success criteria. Second, we provide full-population shortcut and coverage audits, a constructive hidden-structure counterexample, and original-order execution checks. These expose limits of the benchmark as well as limits of simple solvers. Third, we provide replayable exploratory model evidence and a concrete, unfilled confirmation matrix. This is a benchmark-development study, not a new model architecture or a claim of broad model superi-

A. Declared observations



B. Shared source, distinct tasks

Recognize parts	Query relations
Reconstruct	Generate
Complete	Edit
Assemble / disassemble	Detect / repair

Other tasks use their declared symbolic inputs.

C. Evaluation contract



Figure 1. BrickAtlas overview. Real archived views of one two-piece development probe illustrate RGB input; they are not representative of the 9–47-piece casebank. Eight task families share source groups but not necessarily modalities: relations, edits, and plans receive symbolic structures. Layer and symbolic references explicitly add information. The evaluator scores the submitted answer without secretly repairing its geometry or reordering its plan.

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2. Related Work

Brick generation and reconstruction. BrickGPT [6], initially named LegoGPT, combines autoregressive generation with validity checks and physics-aware rollback. BrickNet [4] expands beyond voxel-like towers to human-designed objects and typed connections. Its paper reports 320,808 samples and 9,743 part variants; those units are not directly equivalent to our deduplicated procedural source objects or 25-type catalog. Neither their dataset scale nor their physical scope can be claimed by importing only a small connector annotation subset. MEPNet [10] uses component geometry and stepwise manual images for high-precision pose prediction. Its observation protocol differs from reconstructing an entire opaque assembly. Table 1 summarizes the relevant distinctions without interpreting an unreported capability as absent.

Visual programs and diagnostic reasoning. CSGNet [8] parses shapes into executable constructive solid geometry programs, establishing that visual-to-program prediction is not new. CLEVR [3] uses controlled scenes and functional questions to diagnose compositional reasoning. BrickAtlas extends this diagnostic perspective to shared assembly objects, changed regions, admissible hidden layouts, and execution order. This controlled setting trades semantic realism for inspectable variables; procedural scale alone cannot justify generalization to natural images or arbitrary CAD. BLINK [1] evaluates perceptual tasks including spatial re-

lations and multi-view reasoning with human and specialist baselines. Our complementary target is executable, editable assembly structure; we do not claim that general visual perception is untested.

Presentation bias and training controls. Option-position sensitivity is established in language-model evaluation [11]. Our candidate study therefore does not claim to discover that bias. It compares *actual selected poses*, not answer letters, against a same-order repetition baseline in a geometric domain. We do not infer token-level mechanisms from API outputs. Earlier Qwen3 [7] and SmolVLM [5] LoRA [2] pilots are implementation controls, not proposed architectures or evidence of multi-task transfer.

3. The BrickAtlas Benchmark

3.1. Domain and Source Construction

A part is $p = (i, t, c, x, y, z, r)$: instance ID, catalog type, color, minimum body corner, and quarter-turn yaw $r \in \{0, 1, 2, 3\}$. The vertical axis is Y ; X, Z are horizontal. Horizontal units are 8 mm studs; vertical units are 3.2 mm plate heights. Odd yaw swaps the catalog width and depth. Origins are integer-valued in $[0, 24]^3$, bodies must fit in $[0, 32]^3$, and a submission contains at most 64 parts. Rectangular bricks have height three and plates height one. Nominal validity requires non-overlap and at least one supporting stud for each elevated part. It does not establish force balance, clutch strength, deformation, or robotic grasp feasibility.

We accept 640 unique geometries from each of eight

Resource / method	Input and domain	Main target	Relation to BrickAtlas
BrickGPT [6]	Text; generated brick structures	Stable, buildable generation	Stability-aware generation, not our nominal grid check
BrickNet [4]	Human-designed LDraw; typed connectors	Graph-backed generative programs	Richer part and connector coverage
MEPNet [10]	Stepwise manuals + component geometry	Component pose / executable plan	Strong task-specific baseline after input adaptation
Break and Make [9]	Interactive LTRON inspection	Disassemble and reconstruct	Interactive evidence, unlike fixed-input submissions
CLEVR [3]	Rendered scenes + questions	Compositional visual QA	Diagnostic control, without our assembly-edit contract
BrickAtlas (this work)	Procedural grid; explicit RGB / symbolic arms	Shared-object, typed task evaluation	Nominal geometry; no physical-stability claim

Table 1. Positioning by published task and observation protocol, not a performance comparison. Specialized methods require audited task adapters before any scores can be compared.

Split	Source objects	Task-condition cases
train	2,690	61,870
validation	392	9,016
test_id	758	17,584
test_ood	1,280	29,440
Total	5,120	117,910

Table 2. The ID case count includes 150 isolated catalog questions, separate from the 5,120 assemblies.

seeded construction policies (rendered gallery in the supplement). Tip walk extends a single supporting parent; low fan favors low attachments; shallow terrace limits height; plate network uses thin parts; column field varies independent columns; alternating beams alternates orientation. Two-support bridge constructs paired supports before a span, and enclosed shell places a perimeter and roof around varying interiors. All sources start with a 4-by-8 plate. The generator normalizes minimum X, Z to zero and checks insertion validity and an independently implemented cell certificate.

Geometry is canonicalized after removing colors, translating, and enumerating four global yaw rotations. Exact duplicates within the release and against two earlier releases are rejected. Mirroring is not part of this equivalence. Every task, color condition, or derived prefix retains its source group; a new question is not a new independent object.

3.2. Splits and Measured Coverage

Six policies use a deterministic geometry-hash partition: seven hash buckets for training, one for validation, and two for ID testing. Bridge and shell are entirely policy-held-out. The realized counts are in Table 2. “Policy-held-out” describes the generator split, not absence of shared motifs or proof of learned out-of-distribution generalization.

The new source census measures actual, not nominal,

coverage (Fig. 2). Assemblies contain 9–47 parts and use **19 of the 25 catalog types**. Six types occur only in isolated recognition questions. Piece-count bins contain 1,445 small (≤ 16), 2,141 medium (17–28), and 1,534 large (> 28) objects. These are mechanical bins, not calibrated human difficulty. The policy-by-size matrix exposes missing strata; in particular, policy and size are not independent. A marginal decline with piece count cannot by itself be attributed to sequence length.

A length/prefix Jaccard join at threshold 0.8 on canonical part-position sets is cross-checked over all 8,305,836 cross-split pairs; none reaches the threshold. This result concerns that representation only: canonical orientation can change under perturbation, and the audit does not search for best-aligned subassemblies, semantic duplicates, or pretraining contamination. Existing model-exposed objects remain development evidence in a cumulative exposure register, even when their original dataset split was named “test.”

3.3. Eight Task Families

Each source produces 13 task variants and 23 task-condition cases (Table 3). Reconstruction, completion, and each of three repair variants have three information conditions. Two relation queries, generation, three edits, and two plans use their declared default inputs. Separately, 25 catalog types in six colors yield 150 recognition questions. Thus $5,120 \times 23 + 150 = 117,910$ cases; counting variants without repeated information conditions gives 66,710, not another dataset.

Recognition and relations. Recognition uses isolated isometric/top images and returns type, color, and top-stud count. It is known-catalog identification, not novel-part generalization. Relations supply the full symbolic assembly and two IDs. One sampled pair has direct contact; the other lacks direct contact. The latter can still be connected by a

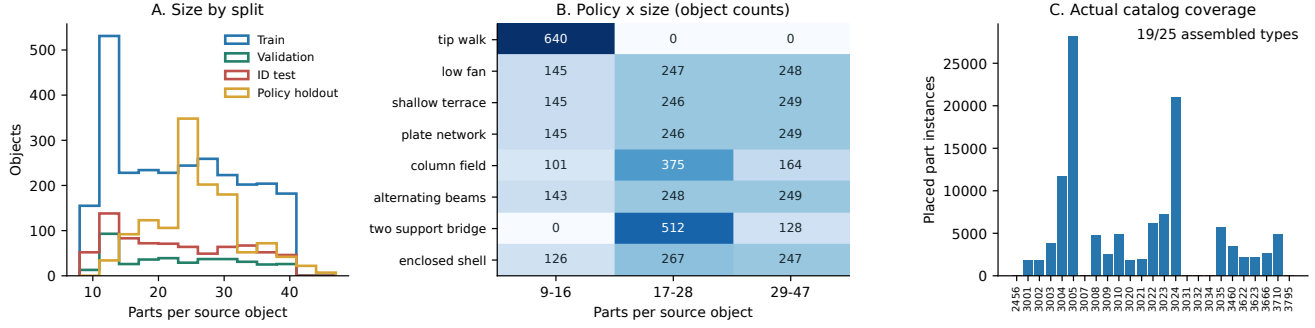


Figure 2. Source-level coverage computed from all 5,120 frozen records. Views and tasks do not contribute extra objects. Zero cells are retained in the policy-by-size matrix; the 25-type catalog exceeds the 19 types actually used in assemblies. Supplementary tables list exact type counts.

Family (variants)	Public input	Required output	Exact task success	Cases
Part recognition	2 isolated RGB views	Type, color, stud count	All three fields correct	150
Relations (2)	Symbolic structure + two IDs	Connection, above, studs, distance	All four fields correct	10,240
Reconstruction	Reference + BOM	Full structure	Condition-specific match + legality	15,360
Generation	Six public constraints	Any satisfying structure	All constraints + legality	5,120
Completion	Current prefix + reference + BOM	Full structure	Match + preserved prefix	15,360
Editing (3)	Current structure + operation	Full edited structure	Exact edit + protected preservation	15,360
Planning (2)	Target structure + direction	Permutation of all IDs	Complete executable order	10,240
Repair (3)	Current + reference + BOM	Structure + faulty current IDs	Match + preservation + localization	46,080

Table 3. Task contract and full-release counts. “Reference” means the declared RGB, layer, or symbolic condition. A structure is a typed, colored pose multiset; arbitrary output IDs are allowed except where a plan or fault localization explicitly refers to current IDs.

longer graph path. Outputs include direct connectivity, vertical ordering, mating-stud count, and shortest undirected contact-graph distance. “Above” means the entire first body starts at or above the second body’s top, not overlap in a projected image.

Reconstruction and constrained generation. Reconstruction returns a complete assembly matching its reference condition and bill of materials (BOM). Generation instead receives exact extents, zero anchoring, a piece budget, minimum color count, connected component count, and a minimum maximum graph degree. Any nominally valid structure satisfying these constraints is correct; similarity to a particular generated witness is not a generation-quality score. The current task is constraint generation, not natural-language semantic shape generation.

Completion, editing, and repair. Completion removes the last $\lfloor n/4 \rfloor$ parts of the valid generation order and asks for the full restored structure while preserving the prefix. It

tests suffix completion, not arbitrary holes. Edits comprise a whole-object quarter-turn, replacement of every instance of a selected color, and removal of one topmost piece. Rotation maps positive X toward positive Z , retains Y , and reanchors minimum X, Z to zero. Repair has no-fault, wrong-color, and one-stud-shift variants; positive faults affect an exposed topmost part, not arbitrary hidden corruption. It requires both a corrected structure and the faulty *current* IDs. Variant labels and changed IDs are evaluator metadata, not ordinary prompt fields.

Assembly and disassembly. The target program is supplied with IDs sorted lexically, preventing the source insertion order from directly serving as the answer. A model returns all IDs once in its chosen order. Only vertical insertion/removal is allowed. These tasks test order under known geometry, not visual reconstruction, motion planning, or grasping.

3.4. Information Conditions

Ordinary RGB provides opaque assembled isometric, top, front, and side views plus BOM. The three axis views look from positive Y , Z , and X , respectively. Cameras use orthographic projections and recenter/rescale the assembly; screen pixels are not absolute grid coordinates. Ordinary success accepts legal BOM-matched alternatives with equal colored positive-axis first-hit grid surfaces. **Layer disclosure** provides an assembled view and one top image per occupied bottom- Y layer with other layers removed; it requires full type/color/pose recovery. **Symbolic reference** supplies the complete target program and also requires full recovery. It is privileged diagnostic information, not a visual score. All three retain applicable preservation and localization requirements. Render recipes and ground-truth records are never appended to ordinary model inputs.

4. Evaluation and Measurement Validity

4.1. Accepted Answers, Not Only Reference Programs

Let T and A be target and answer multisets, and m the number of type/color/pose matches after part-yaw symmetry. Instance IDs do not affect these structural matches. We report

$$P = m/|A|, \quad R = m/|T|, \quad F_1 = 2m/(|A| + |T|). \quad (1)$$

Multiset matching prevents duplicate predictions from increasing recall. BOM F1 removes poses; occupancy IoU removes part boundaries; colored occupancy preserves colors. Edge F1 requires matched endpoints and the correct number of mating studs. These quantities answer different questions and are not averaged into one “intelligence” score.

Let $S^+(A)$ be the colored first-hit cell set along positive X, Y, Z . For ordinary visual cases, the base structural predicate is

$$V(A, T) = \mathbf{1}\{\text{valid}(A)\} \cdot \mathbf{1}\{\text{BOM}(A) = \text{BOM}(T)\} \cdot \mathbf{1}\{S^+(A) = S^+(T)\}. \quad (2)$$

It is a grid-surface proxy, not pixel identity or a proof of observability from arbitrary cameras. Full-reference success is reported separately. For completion, retained source matches are subtracted before computing addition precision/recall/F1. Editing and repair require protected-piece preservation; repair also requires exact faulty-ID localization. False alarms are defined only on no-fault cases.

RQ1: why global F1 can reward doing nothing. For a completion with n target parts and k missing parts, copying the current $n - k$ pieces yields

$$F_1^{\text{copy}} = \frac{2(n - k)}{2n - k}, \quad F_{1, \text{added}}^{\text{copy}} = 0. \quad (3)$$

With $k \approx n/4$, global F1 is already about 6/7. For removing one piece, copying all n original parts instead yields $2(n - 1)/(2n - 1)$, approaching one as assemblies grow. This is a property of the metric, not evidence that the model performed the operation.

RQ2: matching observations need not match relations.

For the declared admissible set $\mathcal{A}(T) = \{A : V(A, T) = 1\}$, a downstream query q has a determined answer only if $q(A)$ is constant over $\mathcal{A}(T)$. Figure 3 gives a constructive counterexample. Two hidden tilings have identical BOM, colored occupancy, and declared first-hit surfaces, but different contact graphs and a different answer to whether two upper pieces share one lower support. Therefore an ordinary accepted reconstruction cannot automatically inherit every relation label from the reference. This is one existence witness, not a prevalence estimate, human judgment, or pixel-equality claim. V2 relation questions avoid this ambiguity by supplying symbolic geometry; extending them to images would require additional observations or set-valued/undetermined answers.

4.2. Executable Order and a Simple Complete Baseline

Plans execute in the submitted order and stop at the first unknown, duplicate, unsupported, colliding, or blocked action. We record the zero-based failure step, legal-prefix fraction, executed coverage, unique submitted coverage, length accuracy, and duplicate-free status. An incomplete but legal prefix fails exact success and has no illegal step. The evaluator never sorts a failed plan into a passing one.

Height-order sufficiency. Within the valid rectangular grid domain, sorting parts by increasing bottom Y gives a complete assembly order. Every supporter lies strictly lower and is already inserted. Any earlier part overlapping the new footprint must end at or below its bottom, otherwise the final structure would overlap; hence no earlier part blocks insertion from above. Equal-height non-overlapping parts can be ordered arbitrarily. Reversing this sequence gives a valid disassembly order. The complete 10,240-case execution audit agrees with this construction. Consequently, failures of a myopic greedy planner demonstrate the importance of order but do not establish hard combinatorial search.

4.3. Denominators, Reproducibility, and Independent Checks

A selection manifest freezes IDs, filters, and hashes before scoring. Unknown/duplicate prediction IDs are rejected. Missing or malformed submissions stay in applicable denominators; null denotes inapplicability, not an omitted failure. A versioned strict primitive-type decoder closes a

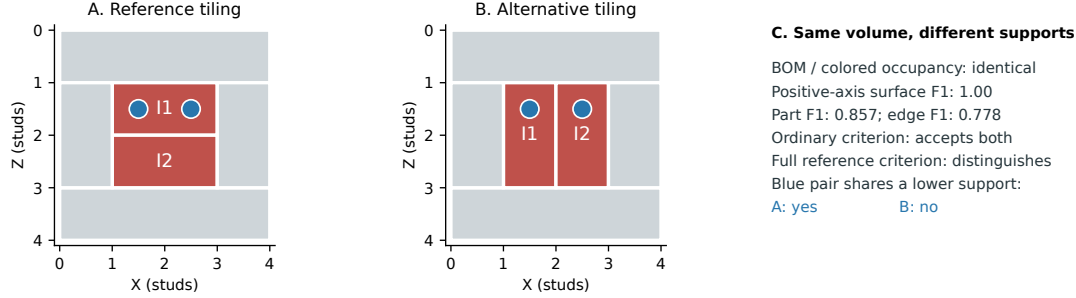


Figure 3. A disclosed cross-section of the independently checked 14-part hidden-layout witness, not an ordinary input render. Red rectangles show internal bricks at bottom $Y = 1$; blue markers project the two query bricks at $Y = 4$ onto that layer. The roof and upper walls are omitted only for explanation. In A, both blue bricks share one red supporter; in B, they do not.

legacy JavaScript coercion gap without rewriting historical results. All 117,910 oracle answers still pass.

Every ground-truth record links public-input hash, source/split, target, protected/changed IDs, contacts, nominal stud mates, metric definitions, and an executable witness. Separate implementations check cell geometry, structural metrics, and relation answers, with mutation/invariance tests for malformed submissions. These reduce coding risk but share catalog and nominal assumptions; they do not replace human calibration. Detailed counts and replay contracts are in the supplement.

Replay distinguishes model output, harness operations, and evaluator diagnostics; unfolding a structure for display does not count as a model plan. Uncertainty uses 1,000 fixed-seed bootstrap resamples of source-group means, keeping derivatives clustered and information conditions separate.

5. Experiments and Findings

5.1. Full-Casebank Algorithmic and Shortcut Audits

The public-input deterministic baseline solves every relation and edit case and all 25,600 privileged symbolic reconstruction, completion, and repair cases. It abstains on unsupported visual and generation inputs; those zeros indicate coverage, not failed visual reasoning. Its immediately-legal greedy planner solves 2,364/5,120 assemblies (46.2%) and all disassemblies. A roof can be placed legally before a lower piece and block its future insertion corridor. Height sorting solves both directions completely. Split-specific results are retained in the supplement; because neither planner is learned, their split contrast is not learned generalization.

We newly evaluate a copy-current predictor on **35,840 cases**, one ordinary completion, three symbolic edits, and three ordinary repairs per source. Its predictor sees only public `current`; it returns that structure unchanged and an empty fault list for repair. Table 4 is a census of the

Copy-current task	Cases	Success	Part F1	Local
complete/suffix	5,120	0.0	87.1	0.0
edit/rotate	5,120	0.0	0.6	0.0
edit/recolor	5,120	0.0	79.0	0.0
edit/remove	5,120	0.0	97.4	0.0
repair/none	5,120	100.0	100.0	100.0
repair/color	5,120	0.0	94.9	0.0
repair/shift	5,120	0.0	94.9	0.0

Table 4. New full-casebank no-operation audit, percentages. Local is addition F1 for completion, exact target success for edits, and localization F1 for repair. Rows share the same 5,120 sources. High global similarity is not successful task execution.

full released population, including training objects, not a new held-out model experiment. The 87.1% completion and 97.4% removal F1 scores confirm Eq. 3. Zero addition F1 or target success correctly reveals that the requested change was not performed. No-fault repair is the intended exception: leaving a correct structure unchanged is the correct response.

5.2. Fixed-Input Visual and Symbolic Model Study

Before calls, we select three held-out objects per policy across available size bins: 24 assemblies, all 13 ordinary/default variants, and 25 separate catalog questions yield 337 cases per model. GPT-4.1 mini and Gemini 2.5 Flash run interleaved at temperature zero with a 2,200-token output cap. All 674 responses, failures, image references, token counts, latencies, and receipts are archived. Table 5 reports actual successes, not a projected main leaderboard. Source-level bootstrap summaries and partial metrics are available in the accompanying evidence.

Both models fail exact ordinary reconstruction, completion, generation, and rotation, while recoloring/removal saturate. This combination of floors and ceilings limits ranking resolution. Six reconstruction controls and one assembly-representation control on one source per policy add 56 Gem-

Task / variant	GPT-4.1 mini	Gemini Flash
parts	8/25	12/25
relations/contact	7/24	7/24
relations/separated	0/24	0/24
reconstruct/full	0/24	0/24
generate/constraints	0/24	0/24
complete/suffix	0/24	0/24
edit/rotate	0/24	0/24
edit/recolor	24/24	24/24
edit/remove	24/24	24/24
plan/assemble	1/24	12/24
plan/disassemble	1/24	7/24
repair/none	21/24	17/24
repair/color	4/24	6/24
repair/shift	0/24	0/24

Table 5. Exploratory v2 one-shot results. Each structural row uses the same 24 objects; recognition uses 25 isolated catalog types. Symbolic default tasks are not visual results.

ini calls. Symbolic references solve 8/8 controlled reconstructions; no-image, mismatched-image, repeat, reflection, and legality-feedback arms remain at zero exact reconstruction. That floor does not establish that the model ignores images. Reflection and validation reuse the identical first answer, and legality feedback is target-blind. Complete control results are in the supplement.

Figure 4 distinguishes observed failure gates. JSON/schema/domain rejection, invalid geometry, and task mismatch are operational labels, not inferred cognitive causes. Ten local SmolVLM-256M base/LoRA jobs yield no positive edit transfer. Base fails JSON parsing on 298 cases and schema on 39, so zero success is not a perception-only finding. Matched-supervision training details and separate historical pilots are in the supplement.

5.3. Coordinate and Output-Protocol Diagnostics

A three-source screen adds GPT-4.1 and Qwen3-VL-32B-Instruct. A nested-prefix follow-up reveals inconsistent origin wording; the supplement documents a separately versioned correction without replacing original scores.

Four unused sources then provide distinct normalized two-piece scenes, gray bases, blue query parts, known identities, and explicit camera/grid conventions. Four models answer six conditions each (96 calls). Full-structure and local-pose success both total 8/16; shorter output alone is not supported as a general remedy. Candidates have equal identities, yaw, and height and differ in legal horizontal position. Only 6/16 paired candidate answers preserve the actual pose after reordering. These four-source probes lack an identical-repeat control, motivating the next experiment.

5.4. RQ3: Candidate Reordering Beyond Repetition

A follow-up freezes 12 new validation sources, two per seen policy, with distinct normalized two-piece geometries. Original, identical repeat, permutation, and no-image conditions yield 192 calls across four models. Correct-option positions and repeat/permutation timing are counterbalanced. For source g , let a_g, a_g^r, a_g^π denote selected *poses*, after mapping option labels back to content. The paired endpoint is

$$D_g = \mathbf{1}[a_g^\pi \neq a_g] - \mathbf{1}[a_g^r \neq a_g]. \quad (4)$$

Mismatch measures inconsistency, not incorrectness; a consistently wrong answer has zero mismatch. Figure 5 reports the paired means and source-bootstrap intervals.

Repeat/permutation mismatches are 2/9 for GPT-4.1 mini, 2/11 for GPT-4.1, 0/6 for Gemini, and 0/9 for Qwen, each out of 12 pairs. Excess mismatch ranges from 0.50 to 0.75. The result supports presentation sensitivity beyond the observed repeat baseline in this development sample. Candidate construction, provider, time, simple geometry, and prior adaptive protocol design remain limitations; the shared 12 sources do not become 48 independent objects by evaluating four models.

A post-hoc renderer-aware analysis-by-synthesis control enumerates legal two-piece scenes from public BOM and candidate poses and matches blue masks in three orthographic views. It solves all 16 original sources across the four-source and 12-source studies and preserves all 16 selected poses after permutation. All 44 image-conditioned answers succeed; those are repeated conditions, not 44 sources. No-image inputs abstain. Replacing views changes 9/15 jointly answered selections, with one abstention. This establishes usable simulator signal in this narrow domain, not general reconstruction or human observability. The method uses the exact renderer and extra compute; its enumeration, thresholds, and resource use are disclosed.

6. Unmeasured Confirmation and Extensions

The preceding evidence is useful for developing a benchmark but does not supply independent confirmation. The supplement provides seven deliberately empty experiment tables: an eight-task main-result layout for new shared sources, paired information/representation controls, candidate invariance and debiasing, difficulty/nuisance factors, external-domain and specialized methods, human calibration, and effective training. The machine-readable plan marks every unmeasured value as null; neither the 24-object pilot nor scores from other papers can substitute for these missing experiments.

Confirmation requires licensed assets, calibrated metrics, effective baselines, and frozen sources, models, inputs, and

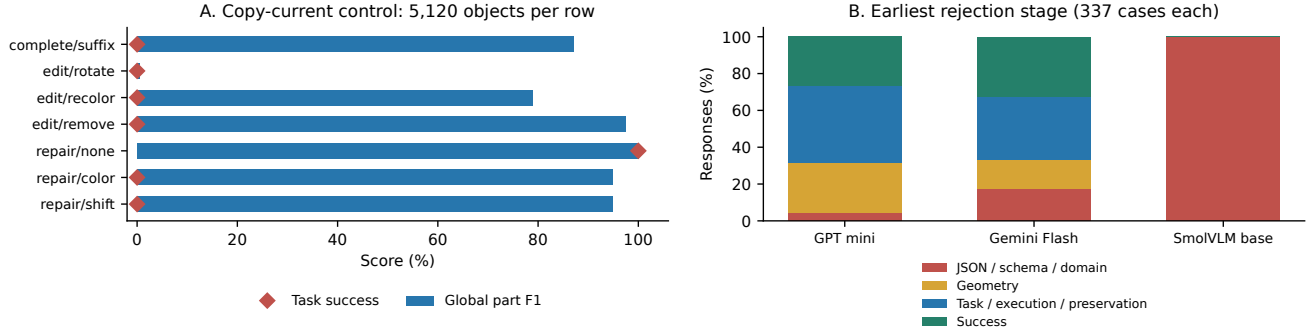


Figure 4. Why a score needs context. Left: copying current geometry often gives high part F1 without task success (full-casebank census). Right: earliest observable rejection gates in the 337-case v2 study; bars mix tasks only to diagnose the interface, not to rank models. The two panels describe different populations and are not paired.

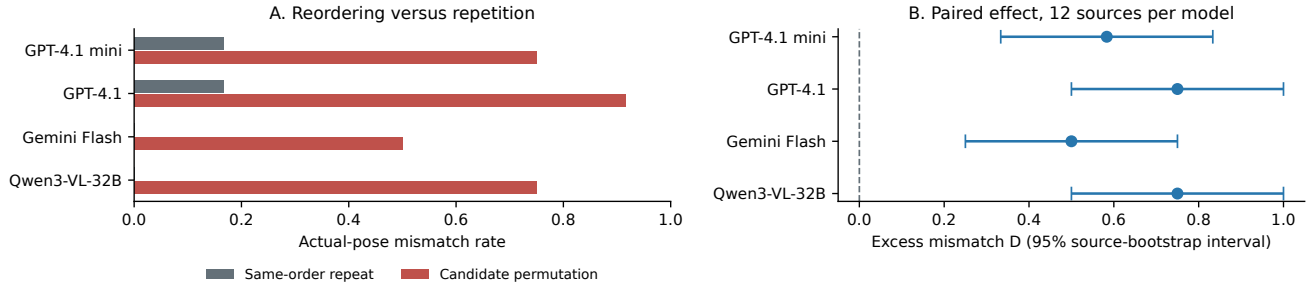


Figure 5. Adaptive development replication on 12 source groups shared across four models. Candidate permutation changes the selected pose more often than identical repetition. The paired excess mismatch intervals describe these selected sources, not broad population coverage or a confirmatory model ranking.

analysis. Specialized methods need matching task information. Sample size follows precision, not desired significance; the supplement specifies precision calculations and missing-pair bounds.

7. Limitations and Conclusion

Scope and remaining validation. All current source assemblies are procedural, share a root plate, and use only 19 assembled types. Recognition covers known parts only. Support is nominal, not physical stability; vertical planning admits a simple complete order. The surface criterion omits some rendered boundary information and does not determine hidden relations. New image-conditioned relation tasks require an ambiguity-aware contract, not automatic inheritance of hidden reference labels. No licensed human-designed structure layer has been integrated.

The review interface contains 96 candidates from eight source objects, but no completed human labels or adjudications. Independent reviewers must measure false acceptance/rejection, disagreements, and unresolved cases against a frozen scoring version. Same-machine, runtime-free replay is not independent second-environment repro-

duction. The current model studies are small, adaptive, and confound some geometric factors. Training losses and engineering test counts do not establish construct validity, positive transfer, or broad ranking reliability.

Release and research practice. Procedural records declare CC0 provenance; other assets retain their own licenses. Credentials and human-review identities are not published. The new census and figures make no API calls; historical resource accounting is in the supplement. This development paper uses the official template, not fabricated submission metadata or an acceptance claim.

BrickAtlas separates visual compatibility, structural recovery, local change, and executable order on shared objects. Its audits expose misleading global similarity, while model diagnostics motivate controlled follow-up. Independent confirmation, human calibration, and external-domain evidence remain necessary for stronger claims.

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